Supplemental Material: Guidance-Coupled Bearing-Only Optical Navigation for Cislunar Transfers

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May 2026

This supplement carries the extended sweeps and detailed sensitivity studies summarized in Section 5.7 of the main paper. All experiments are $n = 1000$ matched-seed Monte Carlo at the canonical operating point ($\sigma_{\text{px}} = 1$ pixel, $q_a = 10^{-14}$, $t_c = 2.0$, estimate-driven active pointing, $\mathbf{P}_0 = \text{diag}(10^{-6}\mathbf{I}_3, 10^{-7}\mathbf{I}_3)$) unless explicitly varied. Section-numbering uses an "S" prefix for clarity. Every figure/table here has a backing $n = 1000$ production CSV under `paper_artifacts/` or `results/mc/` in the project repository; the figure-to-script manifest is `docs/experiment_manifest.md`.

1 Process-Noise Tuning

The process-noise study was performed in two stages. A coarse sweep exposed the failure mode: too little process noise made the filter overconfident, with NEES far above the $\chi^2(6)$ range. A later fine sweep ran 1000 trials at each of 11 candidate q_a values in the final operating regime. Table 1 shows that $q_a = 10^{-9}$ would put the trial-mean NEES inside the diagnostic $\chi^2(6)$ 95% band on 99.4% of trials but inflate the median terminal miss to 1.81×10^{-4} LU ≈ 70.6 km, while the lowest q_a values preserve the 99.1% screening pass rate at the price of a 89% NEES band fraction. The main paper adopts $q_a = 10^{-14}$ as the canonical operating point so the realism sweeps (Sections 3–8) operate inside the same covariance regime.

Table 1: Fine process-noise sweep, 1000 trials per point. "Band" is the fraction of trial-mean NEES values inside the $\chi^2(6)$ 95% interval.

q_a	Pass	Miss med. [LU]	NEES med.	Band
1.0×10^{-12}	0.991	1.58×10^{-4}	5.77	0.893
3.16×10^{-12}	0.991	1.57×10^{-4}	5.74	0.899
1.0×10^{-11}	0.991	1.58×10^{-4}	5.71	0.905
3.16×10^{-11}	0.991	1.58×10^{-4}	5.57	0.925
1.0×10^{-10}	0.991	1.57×10^{-4}	5.22	0.944
3.16×10^{-10}	0.991	1.67×10^{-4}	4.82	0.977
1.0×10^{-9}	0.991	1.81×10^{-4}	4.34	0.994
3.16×10^{-9}	0.989	2.33×10^{-4}	3.92	0.998
1.0×10^{-8}	0.970	2.90×10^{-4}	3.62	0.996
3.16×10^{-8}	0.942	3.37×10^{-4}	3.30	0.995
1.0×10^{-7}	0.904	3.75×10^{-4}	2.97	0.995

2 Sensitivity Studies (legacy $n = 80$)

The pixel-noise and t_c sensitivity uses an earlier $n = 80$ deterministic-seed configuration with active pointing and $q_a = 10^{-9}$. Pixel noise is manageable through $\sigma_{\text{px}} = 3$: the median terminal miss stays below the 10^{-3} LU tolerance, rising from 1.22×10^{-4} LU at $\sigma_{\text{px}} = 0.5$ to 6.95×10^{-4} LU at $\sigma_{\text{px}} = 3$. At $\sigma_{\text{px}} = 5$ the median reaches 1.73×10^{-3} LU and the upper tail is well outside tolerance. Correction time shows a separate geometric effect: at $\sigma_{\text{px}} = 1.5$ the median miss drops from 1.32×10^{-3} LU at $t_c = 0.8$ to 1.55×10^{-4} LU at $t_c = 3.0$.

3 Attitude Uncertainty

The baseline assumes ideal pointing. Zero-mean rotational noise $\theta \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{att}}^2 \mathbf{I}_3)$ is applied to the camera attitude at every measurement step on the truth side only; the filter back-projects through the unperturbed commanded DCM. Sweeping $\sigma_{\text{att}} \in \{0, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1.0\}$ degrees at $n = 1000$ yields the graceful-degradation curve in Table 2. The median terminal miss is statistically indistinguishable from the no-noise baseline through $\sigma_{\text{att}} = 0.03°$ (61.6–64.5 km), inflates to 88 km at 0.1°, and then climbs steeply to 231 km at 0.3° and 1058 km at 1°.

Table 2: Attitude-noise sweep, $n = 1000$ trials per σ_{att}.

σ_{att} [deg]	n	miss med [km]	miss p95 [km]	NIS med.	NEES band
0.000	1000	61.64	228.27	1.94	0.893
0.001	1000	63.34	233.80	1.95	0.910
0.003	1000	63.75	231.42	1.95	0.909
0.010	1000	63.21	232.57	1.96	0.902
0.030	1000	64.49	244.20	2.04	0.893
0.100	1000	88.01	321.89	2.89	0.760
0.300	1000	231.04	836.06	10.50	0.019
1.000	1000	1057.99	3158.46	97.88	0.000

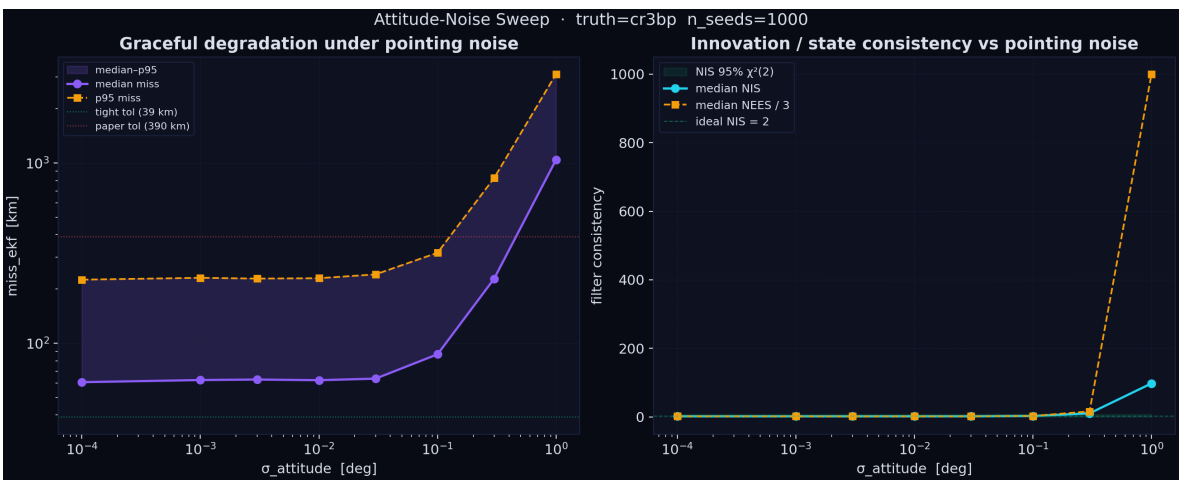

Figure 1: Attitude-noise sensitivity, $n = 1000$ per σ_{att} point. Median and 95th-percentile terminal miss on the left; median Moon-center NIS and median NEES on the right. The performance plateau extends to roughly 0.1° before the filter becomes inconsistent and miss climbs steeply.

4 Cumulative-Parallax Range Observability

A bearing-only filter cannot observe range directly. Range becomes observable through the cumulative angular change of the Moon line-of-sight relative to the trajectory. Sweeping $t_c \in \{0.25, 0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 5.0\}$ at $n = 1000$ trials per t_c yields one (parallax, range error) point per trial. The fit recovers an exponent of -0.57 against cumulative parallax (and -0.71 against the endpoint-to-endpoint quantity); both exponents are negative and order-unity, consistent with the bearing-only intuition that range error decays with accumulated LOS sweep but weaker than a clean inverse law because (a) very-short-arc bins are pre-convergence and (b) long-arc bins flatten at the noise floor.

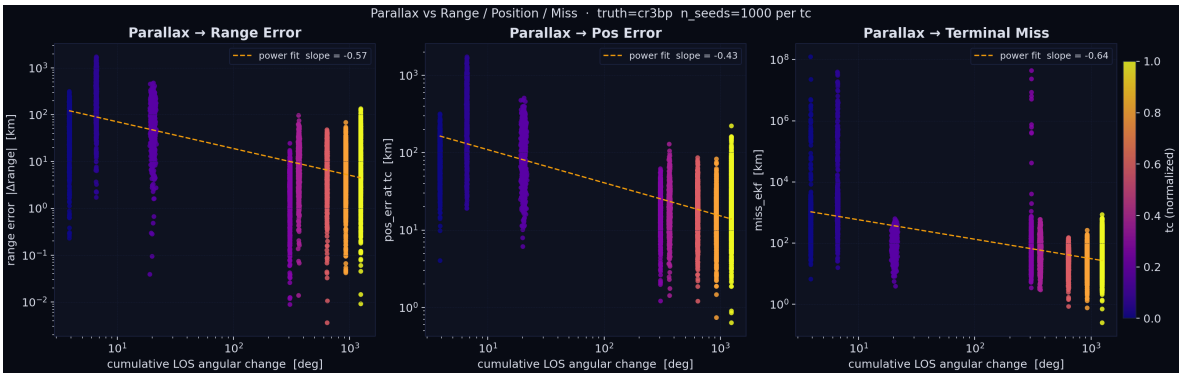

Figure 2: Cumulative-parallax versus range-error fit, $n = 8000$ trials. Empirical exponent -0.57.

5 Filter-Covariance Geometry at Burn

The IEKF position covariance at t_c is anisotropic: the elongated direction aligns with the weak observability mode reported in Section 4 of the main paper. Whether that elongated direction matters for guidance depends on its alignment with the burn sensitivity, captured by the position-to-velocity STM block $\Phi_{rv}(t_f, t_c)$.

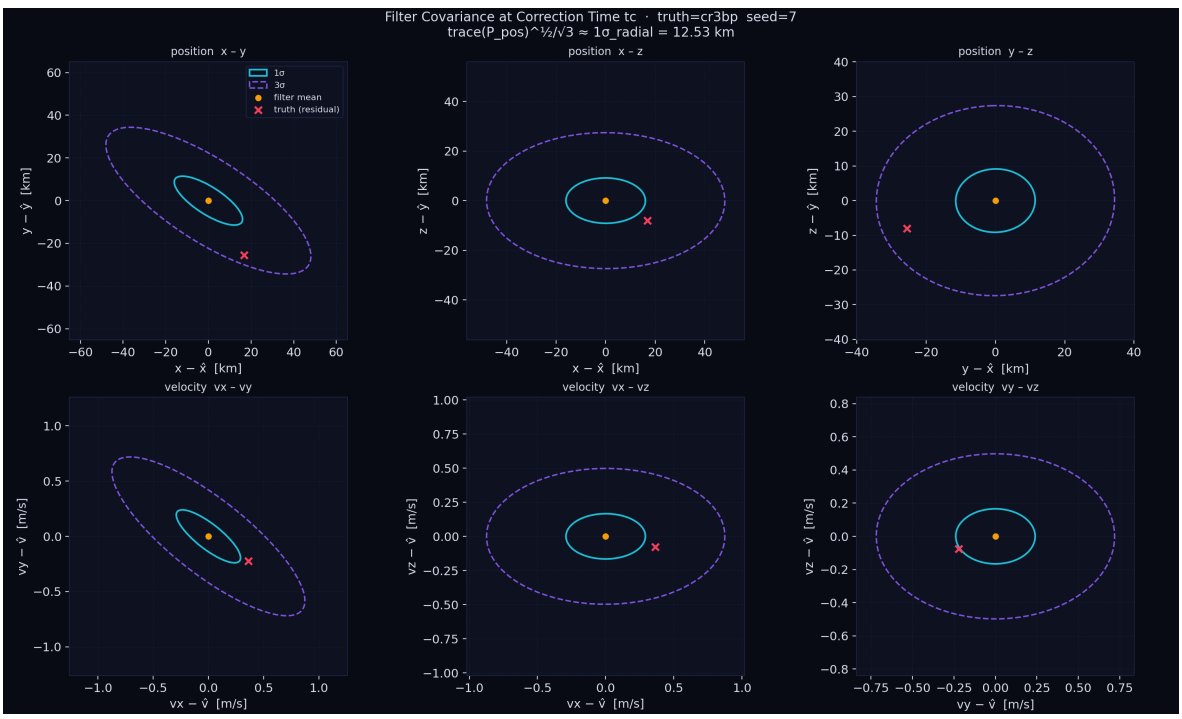

Figure 3: 1σ and 3σ IEKF position-covariance ellipses at the burn epoch t_c in three orthogonal planes, with the realized truth residual overlaid.

6 Initial-Covariance Robustness

The $\mathbf{P}_0$ scale is swept by factors $\{1, 3, 10, 30, 100\}$ at $n = 1000$ trials each. Median miss inflates from 61.6 km at the baseline scale to 116.4 km at $\mathbf{P}_0 \times 100$ (less than $2\times$ degradation across two orders of magnitude in initial covariance), and NIS stays essentially flat at 1.94 across all scales. The headline result is therefore not a tightness artifact: the filter recovers a comparable estimate from a $100\times$ wider initial covariance.

Table 3: Initial-covariance scaling sweep, $n = 1000$ trials per $\mathbf{P}_0$ scale.

$\mathbf{P}_0$ scale	n	miss med. [km]	miss p95 [km]	NEES med.	NIS med.
$\times 1$	1000	61.64	228.27	5.78	1.94
$\times 3$	1000	67.66	221.13	5.18	1.94
$\times 10$	1000	82.74	266.46	5.21	1.94
$\times 30$	1000	101.11	308.13	5.59	1.94
$\times 100$	1000	116.40	361.10	6.33	1.95

7 Multi-Revolution Stability

The estimator is run across longer observation arcs with $t_c = 0.95\, t_f$ and $t_f \in \{6, 9, 12\}$, corresponding to roughly 1.9, 2.9, and 3.8 halo periods near L_1, at $n = 100$ deterministic seeds per t_f. Endpoint position error remains bounded across all arcs (median final position error of 11, 38, and 11 km respectively). The fraction of trial-mean NEES values inside the $\chi^2(6)$ 95% interval is 49.1%, 34.7%, and 26.8% as the arc length grows; this is a consistency degradation, not a divergence, and is consistent with mild over-confidence accumulating over extended arcs.

Figure 4: Multi-revolution position error and NEES envelopes (median and inter-quartile band), $n = 100$ seeds per arc length.

8 Measurement-Delay Sensitivity

Sweeping $\Delta k \in \{0, 1, 2, 5, 10, 20, 50\}$ measurement steps, with the truth-side image generated at the delayed epoch, produces Table 4. A single-step delay ($\Delta t = 0.02$ ND, roughly 2 hours) inflates the median terminal miss from 61.6 km to 931 km—a 15× degradation—while the median NIS climbs only modestly from 1.94 to 2.04 and remains within the diagnostic $\chi^2(2)$ 95% band. The median NEES, in contrast, balloons by eight orders of magnitude on the same single step. This is the strongest in-pipeline evidence that NIS in isolation is not sufficient as a navigation-health metric.

Table 4: Measurement-delay sweep, $n = 1000$ trials per delay step.

Δk	Δt [ND]	miss med [km]	miss p95 [km]	NIS med.	NEES med.
0	0.000	61.64	228.27	1.94	5.78×10^{0}
1	0.020	931.18	1972.39	2.04	9.89×10^{7}
2	0.040	997.21	2708.78	2.37	1.00×10^{9}
5	0.100	1467.17	3571.62	7.59	7.16×10^{9}
10	0.200	11816.04	15825.36	61.29	1.47×10^{10}
20	0.400	913870.41	36942172.06	286.75	7.40×10^{10}
50	1.000	704584.49	27869912.59	160.74	2.53×10^{10}

9 SPICE Point-Mass Truth Stress Test

A 1000-trial CR3BP-versus-SPICE comparison was generated as a model-mismatch stress test. The SPICE arm is a Sun–Earth–Moon point-mass propagator seeded from JPL ephemerides; it is not a full flight-fidelity model with non-spherical gravity, solar radiation pressure, attitude dynamics, or sensor errors. The median position error at t_c remains nearly the same—about 19 km in both runs—and NIS remains close to the expected $\chi^2(2)$ behavior, but the terminal-miss distribution shifts from a median of about 62 km under CR3BP truth to about 183 km under SPICE truth, and the NEES tail grows. Innovation consistency alone therefore does not fully validate model fidelity; state consistency and guidance performance expose errors that measurement residuals can hide.

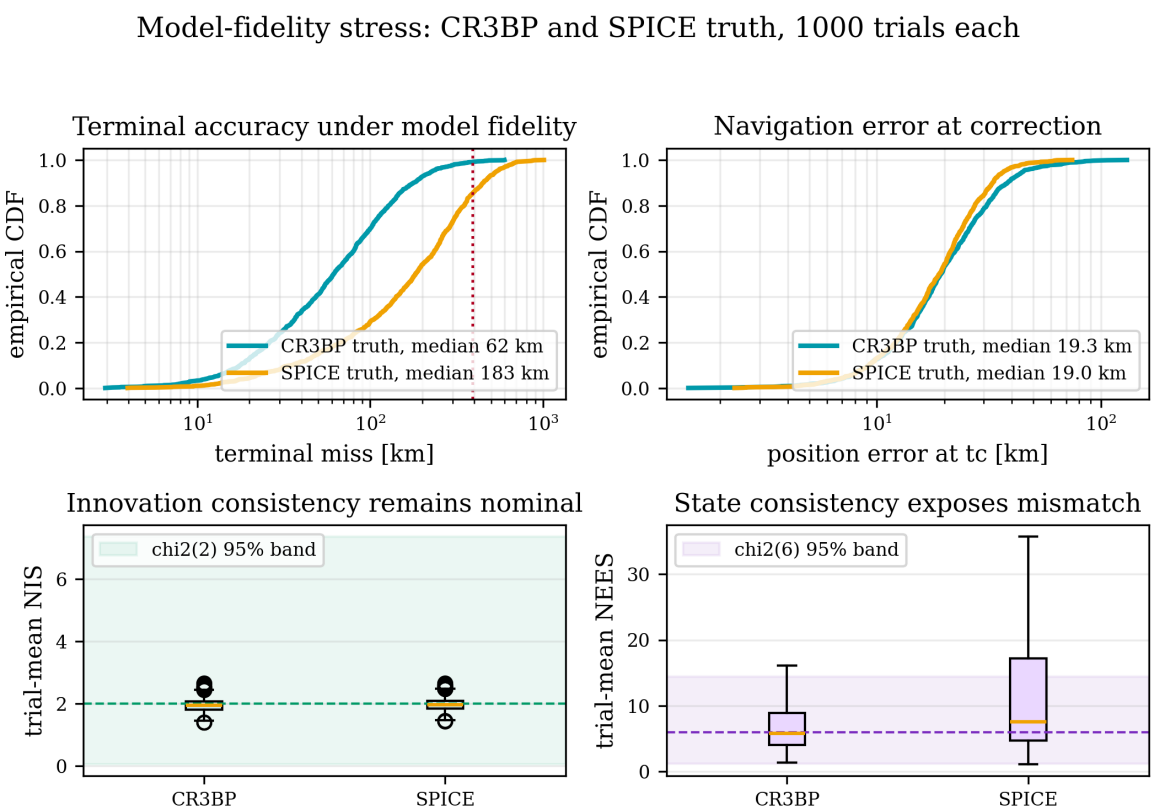

Figure 5: Model-fidelity stress summary for 1000-trial CR3BP and SPICE point-mass truth cases. CR3BP errors are converted to kilometres. NIS remains nominal while NEES and terminal miss reveal the effect of higher-fidelity truth.

10 Synthetic and Catalog-Backed Landmark Configurations

The L1 and L2 landmark sweeps each run three configurations on identical seeds at $n = 1000$ trials per configuration: Moon-center only, landmark-only with the Moon-center disabled, and Moon-plus-landmarks. Median terminal miss drops from 61.6 km (Moon-only) to 26.2 km (landmarks only, L1) and 23.5 km (Moon+landmarks, L1). The L2 catalog-backed numbers (Tycho, Copernicus, Aristarchus, Plato, Kepler, Grimaldi) are within 6% of the L1 numbers across all three configurations, confirming that the gain comes from spatial diversity of the bearing rays rather than the symmetric placement of the L1 layout.

Table 5: L1 (synthetic six-axis) and L2 (catalog six-crater) landmark configurations, $n = 1000$ trials per cell, identical seeds across cells. Pooled NIS is computed over Moon-center plus accepted landmark updates.

Configuration	miss med [km]	miss p95 [km]	pos err [km]	NEES med.	NIS pooled
L1 synthetic six-axis layout					
moon_only	61.64	228.27	19.32	5.78	1.94
landmarks_only	26.17	72.09	7.66	4.60	1.98
moon + landmarks	23.51	69.95	7.15	4.69	1.98
L2 catalog six-crater layout (Tycho/Copernicus/Aristarchus/Plato/Kepler/Grimaldi)					
moon_only	61.64	228.27	19.32	5.78	1.94
landmarks_only	26.94	73.90	7.82	4.59	1.98
moon + landmarks	25.07	72.51	7.37	4.73	1.98

11 Fault and Timing Sensitivity

Table 6 summarizes representative diagnostic cases. The qualitative result is that estimate-tracking and truth-tracking are nearly identical in the nominal case (active pointing itself introduces no significant bias when driven by a good estimate); random dropout and pixel outlier scenarios remain stable in the tested settings; a one-step measurement delay fails as in Section 8.

Table 6: Verification and stress-testing cases on the active-tracking pipeline.

Case	vis. fraction	NIS med.	terminal miss [LU]
Nominal estimate-tracking	0.997	1.715	$\sim 10^{-4}$
Truth-tracking	0.997	1.713	$\sim 10^{-4}$
Dropout	0.960	1.792	5.26×10^{-5}
Pixel outliers	1.000	1.716	7.32×10^{-5}
One-step delay	0.977	1.718	2.96×10^{-3}

12 Compute Budget for Production Drivers

Reviewer-grade reproducibility requires realistic compute expectations. Table 7 reports approximate end-to-end wall times for the major production drivers on an eight-core consumer machine. Per-trial compute is dominated by adaptive-tolerance state-and-STM propagation; the IEKF/EKF updates are sub-millisecond after process-pool startup is amortized. Full paper-results regeneration is under three hours wall, exclusive of the SPICE-point-mass-truth comparison (~ 30 s/trial single-process), which adds roughly thirty minutes more.

13 Modeling Assumptions and Limitations

Table 8 consolidates the modeling assumptions across the paper, indicating for each whether the assumption was stress-tested and where the test appears.

Table 7: Approximate wall-clock cost of regenerating the production data backing each main-paper or supplemental table/figure on an eight-core machine.

driver	output dir	cells $\times n$	wall
`06_monte_carlo.py`	`phase_d_production/`	1×1000	$\sim$30 m
`06f_compare_truth_modes.py`	`phase_d_production_spice/`	1×1000	$\sim$8 h
`06k_p0_sensitivity.py`	`phase_e_P0/`	5×1000	$\sim$1 h
`06l_multi_revolution.py`	`phase_e_multi_rev/`	3×100	$\sim$30 m
`06m_parallax_vs_range.py`	`phase_e_parallax/`	$\sim$8000 pts	$\sim$25 m
`06n_landmarks.py` (L1, L2)	`phase_e_landmarks/`, `phase_e_landmarks_catalog/`	3×1000 each	$\sim$25 m each
`06p_measurement_delay.py`	`phase_e_delay/`	7×1000	$\sim$50 m
`06q_attitude_noise_sweep.py`	`phase_e_attitude_noise/`	7×1000	$\sim$30 m
`06r_landmarks_under_pointing_degradation.py`	`phase_f_landmarks_pointing/`	15×1000	$\sim$3 m
`06s_estimator_ablation.py`	`phase_g_estimator_ablation/`	3×1000	$\sim$5 m
`06t_success_ecdf_central.py`	`phase_h_central_ecdf/`	post-hoc	$<$1 s

Table 8: Modeling assumptions, stress-test status, and what remains out of scope.

Assumption	Baseline	Stress-tested?	Future work
Dynamics	CR3BP estimator; CR3BP or SPICE point-mass truth	Yes: SPICE point-mass truth (§8)	Higher-order non-spherical gravity, SRP
Measurement	Ideal body-center pinhole bearing	Partially: pixel noise, dropout, outliers	Limb fitting, photometric bias, false matches
Active pointing	Ideal estimate-driven pointing	Yes: bias, lag, attitude-noise sweeps (main Tab. 9; §3)	Slew limits, actuator dynamics, closed-loop attitude estimation
Initial covariance	$\mathbf{P}_0$ fixed at the value in main Sec. 3.7	Yes: $\mathbf{P}_0$ scaling §3	Adaptive initialization
Time tagging	Ideal synchronized measurements	Yes: delay sweep §8	Continuous timing bias, jitter, on-board time-tag estimation
Landmarks	L1 (six synthetic) and L2 (six catalog) with known identity	Yes: §10 and main §5.4	L3: image-based detection, catalog association, false-match rejection